

Muzammil Nawaz Khan

Machine Learning Engineer

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SUMMARY

Machine Learning Engineer with 1+ year of production experience shipping classification and prediction systems in Python. Built a claims-classification system (CatBoost, XGBoost, AutoGluon) at 95% accuracy across 42,900+ cases with daily feedback-loop retraining, and a denial-prediction system that cut the denial rate from 3.2% to 2.4%. Experience serving models with FastAPI, containerizing with Docker, and benchmarking deep learning models (3D U-Net vs SegFormer3D, 85%+ Dice). Open to remote work and relocation.

SKILLS

Machine Learning: Python, PyTorch, TensorFlow, Keras, scikit-learn, CatBoost, XGBoost, AutoGluon, Hugging Face Transformers, Pandas, NumPy

Computer Vision: OpenCV, U-Net / 3D U-Net, SegFormer / SegFormer3D, YOLO, GradCAM, medical image segmentation

Generative AI & Agents: OpenAI API, Anthropic API, LLMs, prompt engineering, structured outputs, RAG, LLM-as-judge evaluation, LangChain, LangGraph, CrewAI, AutoGen, Model Context Protocol (MCP), n8n, ChromaDB, FAISS, custom HNSW, BM25, Reciprocal Rank Fusion

Backend & Infrastructure: FastAPI, Django / DRF, REST APIs, Docker, Kubernetes, Git/GitHub, CI/CD

Languages: Python, SQL, Rust, C++, JavaScript/TypeScript

Domain: Healthcare IT, medical imaging, AI training data

EXPERIENCE

Junior AI Engineer, CareCloud (MTBC) Jul 2025 – Aug 2026
Islamabad, Pakistan

- Delivered a production claims-classification system (CatBoost, XGBoost, AutoGluon) at 95% parent-model accuracy and 92%+ category-level accuracy across 42,900+ cases, with daily feedback-loop retraining and multi-month tracking of accuracy and stability.
- Built a Python denial-prediction and resubmission automation system with a rule-based First-Time Pass Rate (FTPR) module comparing rejected vs corrected/paid claims by provider, payer, CPT and ICD code. Cut resubmission wait from 2 weeks to 3 days and reduced the denial rate from 3.2% to 2.4%.
- Designed an n8n workflow acting as a Model Context Protocol (MCP) server for conversation state, backend prompts and call routing, wrapping APIs as tool calls. Cut inference costs 35% through prompt caching.
- Engineered Python pipelines that pull Hikvision NVR logs, clean them and load them to a database for HR attendance analytics; built an internal FastAPI EDI Viewer for inspecting raw healthcare claim data.
- Layered rule-based checks with an LLM validation agent that generates root-cause analysis and corrective-action recommendations for each flagged case.

AI/ML Research Intern, Risetech Jul 2024 – Aug 2024
Rawalpindi / Islamabad, Pakistan

- Segmented brain tumor regions from multi-modal MRI (T1-CE, T2, FLAIR) on the BraTS20 dataset.
- Benchmarked 3D U-Net against SegFormer3D and selected SegFormer3D (85%+ Dice). Used GradCAM for explainability and FastAPI for serving.
- The work became the Neurofusion final year project: 1st Place at COMPPEC 2025 and a Departmental Silver Medal.

Software / AI Engineer, Suidhaga Mar 2022 – Nov 2025
Remote, part-time alongside studies

- Applied data analysis and data science to business data; later built ARIMA-family forecasting models to predict product prices.
- Progressed through the full engineering stack over 3+ years for an e-commerce business: built frontend UI, moved to Django-focused backend development, then owned full-stack web features end to end.

Freelance AI Engineer, Independent Sep 2022 – Present
Remote

- Delivered AI development projects for clients, including model training and applied AI application builds.

- Evaluate and rank AI model responses to improve accuracy in different contexts.
- Test models for inaccuracies and bias, validating their applicability in the target domains.

PROJECTS

Freight Rates Predictor (Python, scikit-learn): End-to-end ML pipeline on spatial, load, market-index and temporal features.
[GitHub: github.com/muzammil5539/Machine-Learning-Engineer-Assessment-Ena-Spotter](https://github.com/muzammil5539/Machine-Learning-Engineer-Assessment-Ena-Spotter)

Verifiable Agent Kernel (VAK) (Rust, Python, WASM, Cedar/ABAC): Deterministic control plane for LLM agents: least-privilege ABAC, WASM-sandboxed tools, Z3 formal verification before execution, 3-tier memory (Merkle DAG), multi-agent debate with quadratic-voting consensus, and a PRM fine-tuning toolkit. v1.0 with 80%+ test coverage; CI/CD on Docker and Kubernetes; LangChain/AutoGPT/MCP adapters. [GitHub: github.com/muzammil5539/Verifiable-Agent-Kernel-VAK-](https://github.com/muzammil5539/Verifiable-Agent-Kernel-VAK-)

RAG Custom Engine (Pure Python, GPT-4o-mini): From-scratch hybrid retrieval with a custom HNSW store, Okapi BM25 and Reciprocal Rank Fusion, plus a 3-stage Self-RAG gate (retrieval decision, relevance grading, hallucination check) and contextual compression. About 20% precision gain over dense-vector baselines.

Computer Vision Suite (Python, OpenCV): Retinal vessel segmentation (multi-level thresholding, morphology), skin lesion segmentation (connected component labeling, IoU), optical Braille recognition (contour analysis), license plate localization (edge detection), optic disc localisation.

EDUCATION

Bachelor of Computer Engineering, National University of Sciences and Technology (NUST) Nov 2021 – Jul 2025
Islamabad, Pakistan

CGPA 3.25. Final year project: Neurofusion, 3D brain tumor segmentation (SegFormer3D, BraTS20, 85%+ Dice); 1st Place COMPPEC 2025, Departmental Silver Medal.

F.Sc. Pre-Engineering, Ghazali Premier College Jun 2018 – Aug 2020
Lahore, Pakistan

CERTIFICATIONS

- IBM AI Engineering Professional Certificate
- IBM Deep Learning with PyTorch, Keras and TensorFlow Professional Certificate
- Generative AI Engineering with LLMs Specialization (IBM)
- Large Language Model Operations (LLMOps) Specialization, Duke University
- IBM Generative AI Engineering Professional Certificate
- IBM RAG and Agentic AI Professional Certificate

HONORS & AWARDS

1st Place, COMPPEC 2025 | Departmental Silver Medal